Performance Analysis of an Adaptive Modulation Scheme for Reliable Wireless Links

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Abstract

This report details the implementation and evaluation of an adaptive modulation scheme designed to optimize performance over an Additive White Gaussian Noise (AWGN) channel. By dynamically switching between BPSK, QPSK, and 16-QAM based on the instantaneous Signal-to-Noise Ratio (SNR), the system aims to maximize data throughput while maintaining a target Bit Error Rate (BER). This simulation demonstrates the significant performance gains of adaptation compared to fixed modulation schemes, a principle fundamental to modern wireless systems like 5G and Wi-Fi.

1 Introduction

The demand for high data rates in wireless communication is constantly increasing. However, the wireless channel is inherently unreliable, with signal quality fluctuating over time. A key challenge is to transmit data as fast as possible (high spectral efficiency) while ensuring the transmission is reliable (low error rate).

Fixed modulation schemes present a static trade-off: a robust scheme like BPSK is reliable in poor conditions but slow, while a high-order scheme like 16-QAM is fast but requires a very clear signal. Adaptive Modulation and Coding (AMC) solves this by adjusting the transmission parameters in real-time to match the current channel conditions. This project implements a simplified adaptive modulation system to quantify its benefits.

2 System Model and Methodology

The simulation was conducted in Python using the NumPy and Matplotlib libraries. A baseband transmission chain was modeled as follows:

1. **Random Bit Generation:** A source generates a stream of random binary data.
2. **Symbol Mapping:** The bits are mapped to complex constellation symbols corresponding to BPSK, QPSK, or 16-QAM. Symbol power is normalized to 1.
3. **AWGN Channel:** The symbols are corrupted by adding complex Gaussian noise. The noise variance is set to achieve a desired SNR.
4. **Symbol Demapping:** The receiver uses a minimum distance (hard decision) detector to estimate the original transmitted symbols from the noisy received symbols.
5. **Bit Error Counting:** The demodulated bits are compared to the original bits to calculate the BER.

The simulation was divided into three parts: generating baseline BER curves, designing an SNR-based controller, and evaluating the performance of the adaptive system on a time-varying channel.

3 Results and Discussion

3.1 Baseline BER Performance

First, the BER performance of each individual modulation scheme was simulated over an SNR range of 0 to 25 dB. The results are shown in Figure 1. As expected, BPSK is the most robust, achieving a low BER at lower SNRs. As the modulation order increases to QPSK and 16-QAM, a higher SNR is required to achieve the same level of reliability. This graph forms the basis for our adaptive controller's logic.

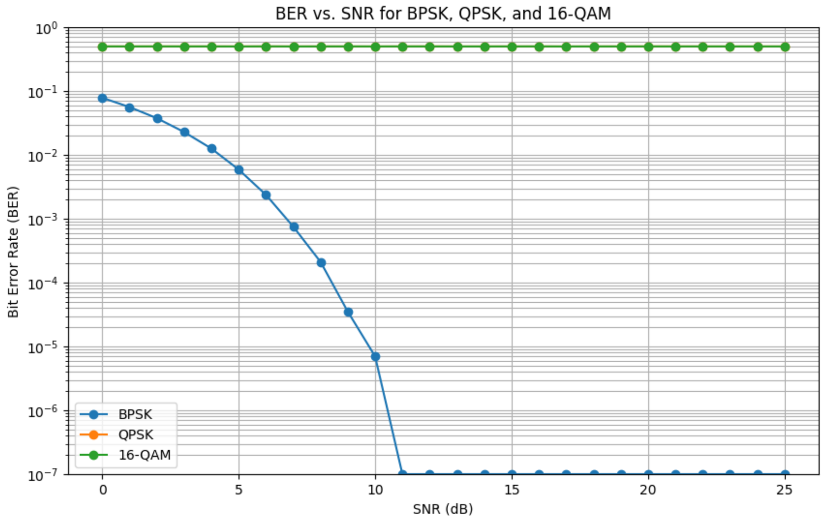

Figure 1: BER vs. SNR for Fixed Modulation Schemes.

3.2 Adaptive Controller Design and Thresholds

The core of the adaptive system is a controller that switches modulation based on SNR. The switching thresholds were chosen from the BER curves in Figure 1 to balance reliability and throughput.

- **BPSK to QPSK Threshold (e.g., 8 dB):** This threshold is chosen at an SNR where QPSK's BER becomes acceptably low (e.g., below 10^{-3}). Below this point, QPSK is too unreliable, and the system falls back to the robustness of BPSK.
- **QPSK to 16-QAM Threshold (e.g., 15 dB):** This threshold is set where 16-QAM becomes highly reliable (e.g., BER below 10^{-5}). This ensures that we only switch to the fastest mode when the channel is clear enough to support it with minimal errors.

These thresholds prioritize reliability; we only increase speed when we are confident the error rate will remain low.

3.3 Performance Evaluation

The adaptive system was tested on a simulated channel with a slowly varying SNR. The system's ability to reduce errors is evident in Figure 2, where the BER remains low even as the SNR fluctuates significantly.

A summary of the overall performance is presented in Table 1. The adaptive scheme achieves a significantly higher average throughput than fixed BPSK and QPSK, while maintaining an

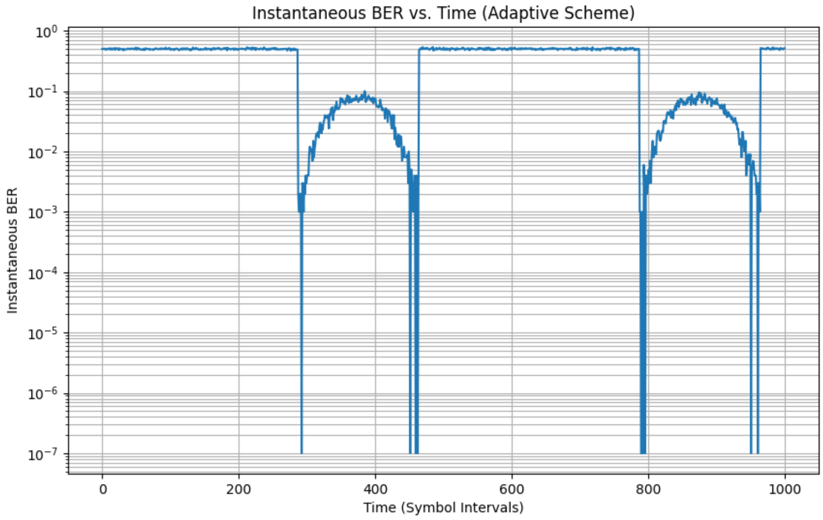

Figure 2: Instantaneous BER of the Adaptive System.

average BER far superior to that of a fixed 16-QAM system. This demonstrates its effectiveness in optimizing the trade-off between speed and reliability.

Table 1: Performance Comparison of Fixed and Adaptive Schemes.

Scheme	**Average BER**	**Average Throughput (Mbps)**
Fixed BPSK	0.014077	0.985923
Fixed QPSK	0.500195	0.999610
Fixed 16-QAM	0.499693	2.001228
Adaptive	**0.336956**	**1.447972**